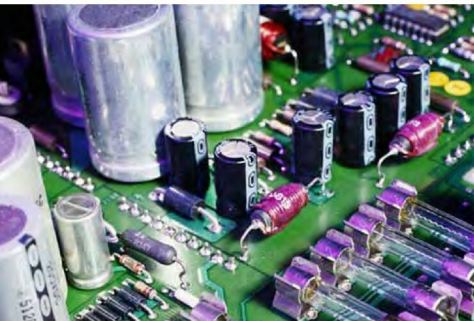


SYSTEM-LEVEL DESIGN

Some common design practices for a better overall system architecture that will eventually lead to a high-quality successful product



■ **ASEEM VASUDEV**

Full system design is a process of integrating the two fundamental system components—hardware and software. The hardware comprises a printed circuit application (PCA) board, which has the main system and, in some cases, multiple daughter boards acting as sub-systems. On the other hand, the software combines low-level drivers with software application or several applications integrated with an operating system (OS).

Complete system validation and qualification are important milestones of any system design. Unless the entire system is tested and qualified, it can't be guaranteed to function under various operating conditions. System validation and qualification in themselves

are very broad fields.

Here we mainly focus on hardware and software components used in developing a system and some common design practices for a better overall system architecture that eventually leads to a high-quality successful product.

System design planning

System cost and design time estimation and planning are the foremost activities to start with, especially when it comes to designing products for the consumer market. The features or the enhancements proposed have to be competitive compared to the other similar products in the market—'value-for-money' often becomes the keyword in consumer market segment.

The hardware design and software application development can be done

simultaneously. To a large extent, the software application can be debugged and validated on code simulators provided by various chip manufacturers for their microprocessors and digital signal processors (DSPs). It is important to ensure that hardware evaluation platforms for such processors and DSPs are made available during early phases of the software development process. Application development and debugging tasks can become much easier and faster with evaluation platforms.

Evaluation platforms are usually electronic boards that have most of the system interfaces around the processor. These boards usually have the most commonly used and recommended peripheral devices interfaced with the central processor.

When customised prototype boards are ready, the application can be migrated to these early prototypes for application-level customisation, further development, and system-level validation and qualification.

PCA board design

From a purely technical standpoint, the PCA board design process involves logical board design, electrical board design and physical board design. The three specialised development teams have to often depend on each other's inputs and interact very often.

The logical design mainly focuses on developing circuit designs after selecting the appropriate electronic/electrical components, schematic design using tools, and generation of net-list and bill-of-material (BOM).

Electrical design involves running signal integrity (SI) checks on various design topologies being used in the